

A Technical Introduction to Principal component analysis

Section 1

Directional component analysis Directional component analysis (DCA) is a method used in the atmospheric sciences for analysing multivariate datasets. Like PCA, it allows for dimension reduction, improved visualization and improved interpretability of large data-sets. Also like PCA, it is based on a covariance matrix derived from the input dataset. The difference between PCA and DCA is that DCA additionally requires the input of a vector direction, referred to as the impact. Whereas PCA maximises explained variance, DCA maximises probability density given impact. The motivation for DCA is to find components of a multivariate dataset that are both likely (measured using probability density) and important (measured using the impact). DCA has been used to find the most likely and most serious heat-wave patterns in weather prediction ensembles, and the most likely and most impactful changes in rainfall due to climate change.

Section 2

Network component analysis Given a matrix E , it tries to decompose it into two matrices such that $E = AP$. A key difference from techniques such as PCA and ICA is that some of the entries of A are constrained to be 0. Here P is termed the regulatory layer. While in general such a decomposition can have multiple solutions, they prove that if the following conditions are satisfied:

Section 3

Market research and indexes of attitude Market research has been an extensive user of PCA. It is used to develop customer satisfaction or customer loyalty scores for products, and with clustering, to develop market segments that may be targeted with advertising campaigns, in much the same way as factorial ecology will locate geographical areas with similar characteristics. PCA rapidly transforms large amounts of data into smaller, easier-to-digest variables that can be more rapidly and readily analyzed. In any consumer questionnaire, there are series of questions designed to elicit consumer attitudes, and principal components seek out latent variables underlying these attitudes. For example, the Oxford Internet Survey in 2013 asked 2000 people about their attitudes and beliefs, and from these analysts extracted four principal component dimensions, which they identified as 'escape', 'social networking', 'efficiency', and 'problem creating'. Another example from Joe Flood in 2008 extracted an attitudinal index toward housing from 28 attitude questions in a national survey of 2697 households in Australia. The first principal component represented a general attitude toward

property and home ownership. The index, or the attitude questions it embodied, could be fed into a General Linear Model of tenure choice. The strongest determinant of private renting by far was the attitude index, rather than income, marital status or household type.

Section 4

Find the empirical mean along each column $j = 1, \dots, p$. Place the calculated mean values into an empirical mean vector u of dimensions $p \times 1$. $u_j = \frac{1}{n} \sum_{i=1}^n X_{ij}$

Section 5

Such dimensionality reduction can be a very useful step for visualising and processing high-dimensional datasets, while still retaining as much of the variance in the dataset as possible. For example, selecting $L = 2$ and keeping only the first two principal components finds the two-dimensional plane through the high-dimensional dataset in which the data are most spread out, so if the data contains clusters these too may be most spread out, and therefore most visible to be plotted out in a two-dimensional diagram; whereas if two directions through the data (or two of the original variables) are chosen at random, the clusters may be much less spread apart from each other, and may in fact be much more likely to substantially overlay each other, making them indistinguishable. Similarly, in regression analysis, the larger the number of explanatory variables allowed, the greater is the chance of overfitting the model, producing conclusions that fail to generalise to other datasets. One approach, especially when there are strong correlations between different possible explanatory variables, is to reduce them to a few principal components and then run the regression against them, a method called principal component regression. Dimensionality reduction may also be appropriate when the variables in a dataset are noisy. If each column of the dataset contains independent identically distributed Gaussian noise, then the columns of T will also contain similarly identically distributed Gaussian noise (such a distribution is invariant under the effects of the matrix W , which can be thought of as a high-dimensional rotation of the co-ordinate axes). However, with more of the total variance concentrated in the first few principal components compared to the same noise variance, the proportionate effect of the noise is less—the first few components achieve a higher signal-to-noise ratio. PCA thus can have the effect of concentrating much of the signal into the first few principal components, which can usefully be captured by dimensionality reduction; while the later principal components may be dominated by noise, and so disposed of without great loss. If the dataset is not too large, the significance of the principal components can be tested using parametric bootstrap, as an aid in determining how many principal components to retain.

Section 6

This power iteration algorithm simply calculates the vector $XT(Xr)$, normalizes, and places the result back in r . The eigenvalue is approximated by $r^T (XTX) r$, which is the Rayleigh quotient on the unit vector r for the covariance matrix XTX . If the largest singular value is well separated from the next largest one, the vector r gets close to the first principal component of X within the number of iterations c , which is small relative to p , at the total cost $2cnp$. The power iteration convergence can be accelerated without noticeably sacrificing the small cost per iteration using more advanced matrix-free methods, such as the Lanczos algorithm or the Locally Optimal Block Preconditioned Conjugate Gradient (LOBPCG) method. Subsequent principal components can be computed one-by-one via deflation or simultaneously as a block. In the former approach, imprecisions in already computed approximate principal components additively affect the accuracy of the subsequently computed principal components, thus increasing the error with every new computation. The latter approach in the block power method replaces single-vectors r and s with block-vectors, matrices R and S . Every column of R approximates one of the leading principal components, while all columns are iterated simultaneously. The main calculation is evaluation of the product $XT(XR)$. Implemented, for example, in LOBPCG, efficient blocking eliminates the accumulation of the errors, allows using high-level BLAS matrix-matrix product functions, and typically leads to faster convergence, compared to the single-vector one-by-one technique.